

CHAMPAIGN 9 SW, IL, USA

WMO: 724360

Lat: 40.0528N

Lon: 88.3729W

Elev: 213

StdP: 98.79

Time Zone: -6.00 (NAC)

Period: 03-19

WBAN: 54808

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.9	-15.7	-22.2	0.5	-19.1	-19.5	0.7	-15.8	10.8	-3.8	9.8	-5.2	2.4	N/A	0.454

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	32.4	24.2	31.2	23.9	30.0	23.2	26.3	30.9	25.3	29.7	24.4	28.5	3.1	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.0	20.6	29.7	24.0	19.4	28.3	23.1	18.3	27.1	82.9	30.8	78.6	29.6	74.8	28.6	29.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.2	8.0	7.0	DB	-23.6	33.8	3.7	1.5	-26.3	34.9	-28.5	35.8	-30.5	36.6	-33.2	37.7	(4)
(5)			WB	-22.8	28.0	3.9	0.7	-25.6	28.5	-27.9	28.9	-30.1	29.4	-32.9	29.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.8	-3.8	-2.3	5.0	11.2	17.3	21.8	22.7	21.7	19.0	12.0	5.1	-0.8	(6)	
	DBStd	10.77	6.92	6.68	6.30	5.12	4.84	3.23	3.00	2.91	4.04	5.25	5.48	5.86	(7)	
	HDD10.0	1546	428	346	181	46	4	0	0	0	1	38	164	338	(8)	
	HDD18.3	3229	686	577	413	217	80	8	3	6	40	207	398	594	(9)	
	CDD10.0	1844	1	2	27	83	231	354	394	364	269	101	17	2	(10)	
	CDD18.3	486	0	0	1	4	48	112	138	112	59	11	0	0	(11)	
	CDH23.3	4563	0	0	12	74	457	1024	1276	1001	600	118	1	0	(12)	
	CDH26.7	1442	0	0	1	9	127	332	436	304	207	26	0	0	(13)	
(14) Wind	WSAvg	3.0	4.0	3.9	3.8	3.9	3.0	2.2	1.7	1.6	1.9	2.8	3.2	3.6	(14)	
(15) Precipitation	PrecAvg	965	48	49	66	103	102	113	112	80	83	72	68	65	(15)	
	PrecMax	1264	169	117	167	212	149	217	202	151	185	190	128	176	(16)	
	PrecMin	808	6	10	22	46	39	37	58	9	18	15	14	30	(17)	
	PrecStd	122	37	29	33	46	36	49	39	37	49	43	35	34	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.4	17.7	25.1	28.0	31.7	33.2	34.6	33.2	33.5	29.4	21.8	16.5	(19)	
		MCWB	13.4	13.4	17.3	17.4	21.7	24.0	26.3	24.5	22.9	21.0	15.3	14.3	(20)	
	2%	DB	11.7	13.8	21.2	25.5	29.8	31.6	32.2	31.4	31.3	26.7	18.7	12.8	(21)	
		MCWB	10.2	11.8	15.7	16.8	21.2	23.8	25.5	24.0	22.7	18.9	13.8	11.3	(22)	
	5%	DB	8.7	10.2	17.7	23.3	28.1	30.3	30.7	29.8	29.2	24.0	16.3	9.8	(23)	
		MCWB	7.0	8.0	13.2	16.0	20.0	23.2	24.9	23.8	22.0	16.9	13.4	8.2	(24)	
	10%	DB	5.2	6.8	15.0	20.7	26.1	28.7	29.2	28.4	26.9	21.1	13.8	7.1	(25)	
		MCWB	4.0	4.9	11.8	14.3	19.3	22.1	23.8	22.6	20.2	16.7	10.5	5.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.9	15.2	18.4	19.6	23.5	26.6	27.8	26.9	25.0	22.6	17.4	14.9	(27)	
		MCDB	14.2	17.5	24.8	24.3	29.3	31.3	32.2	31.2	30.7	27.8	20.0	16.3	(28)	
	2%	WB	10.1	12.7	15.9	18.0	22.5	25.3	26.7	25.6	23.9	20.6	15.1	12.2	(29)	
		MCDB	11.0	14.4	20.2	22.6	28.3	29.9	31.4	29.9	29.5	24.0	17.5	13.5	(30)	
	5%	WB	6.9	9.7	14.0	16.7	21.4	24.2	25.7	24.5	22.8	19.0	12.9	8.8	(31)	
		MCDB	8.2	11.8	16.8	20.9	26.7	28.6	30.1	28.3	27.8	21.5	15.6	10.0	(32)	
	10%	WB	3.7	6.0	11.6	15.3	20.4	23.3	24.9	23.5	21.6	17.1	10.3	6.4	(33)	
		MCDB	4.9	8.0	14.1	19.4	25.1	27.8	28.8	27.0	25.5	20.9	12.9	7.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.6	9.3	10.6	12.4	11.6	11.4	11.0	11.8	13.0	12.4	10.3	7.8	(35)	
		MCDBR	11.6	12.9	14.4	15.7	13.7	12.8	12.0	13.0	14.8	15.3	13.5	10.9	(36)	
	5% WB	MCWBR	10.0	10.1	9.1	8.9	6.4	6.4	6.3	7.0	6.4	8.0	9.7	9.2	(37)	
		MCWBR	11.3	12.6	13.2	12.9	11.8	10.6	11.0	10.7	12.2	11.6	12.5	10.2	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.302	0.316	0.354	0.395	0.436	0.441	0.460	0.457	0.415	0.362	0.315	0.298	(40)	
		taud	2.435	2.386	2.364	2.254	2.192	2.223	2.151	2.148	2.245	2.393	2.483	2.459	(41)	
	Ebn at Noon		865	900	899	879	846	840	819	810	822	835	843	840	(42)	
		Edh at Noon	80	97	111	133	144	140	149	146	124	96	76	73	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.79	2.66	3.65	4.76	5.44	6.24	6.16	5.48	4.62	3.18	2.12	1.51	(44)	
	RadStd		0.20	0.26	0.29	0.34	0.46	0.40	0.34	0.34	0.34	0.34	0.20	0.19	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page